

INNER PRODUCT SPACES AND DIAGONALIZATION

3.1 Dot products and inner products

Definition 3.1.1. An **inner product** on a real vector space V is a function that associates a real number $\langle \mathbf{x}, \mathbf{y} \rangle$ to each pair of vectors $\mathbf{x}$ and $\mathbf{y}$ in V in such a way that the following rules are satisfied: For any vectors $\mathbf{x}, \mathbf{y}$ and $\mathbf{z}$ in V and any scalar k in $\mathbb{R}$,

- (1) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ *(symmetry),*
- (2) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$ *(additivity),*
- (3) $\langle k\mathbf{x}, \mathbf{y} \rangle = k\langle \mathbf{x}, \mathbf{y} \rangle$ *(homogeneity),*
- (4) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, and $\langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$ *(positive definiteness).*

A pair $(V, \langle \cdot, \cdot \rangle)$ of a (real) vector space V and an inner product $\langle \cdot, \cdot \rangle$ is called a **(real) inner product space**. In particular, the pair $(\mathbb{R}^n, \cdot)$ is called the **Euclidean n -space**.

Note that by symmetry (1), additivity (2) and homogeneity (3) also hold for the second variable: i.e.,

- (2') $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$,
- (3') $\langle \mathbf{x}, k\mathbf{y} \rangle = k\langle \mathbf{x}, \mathbf{y} \rangle$.

It is easy to show that $\langle \mathbf{0}, \mathbf{y} \rangle = 0\langle \mathbf{0}, \mathbf{y} \rangle = 0$ and also $\langle \mathbf{x}, \mathbf{0} \rangle = 0$.

Remark 3.1.2. In the above definition, the rules (2) and (3) mean that the inner product is linear for the first variable; and the rules (2') and (3') above mean that the inner product is also linear for the second variable. In this sense, the inner product is called **bilinear**.

If V is a vector space over $\mathbb{C}$, then in the definition of the inner product, we have conjugate symmetry instead of symmetry. That is,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}.$$

Consequently, property (3) will give

$$\langle \mathbf{x}, k\mathbf{y} \rangle = \overline{\langle k\mathbf{y}, \mathbf{x} \rangle} = \overline{k \langle \mathbf{y}, \mathbf{x} \rangle} = \bar{k} \langle \mathbf{x}, \mathbf{y} \rangle, \quad \forall k \in \mathbb{C}.$$

Then we say that the inner product is **conjugate linear**.

Example 3.1.3 (*Non-Euclidean inner product on $\mathbb{R}^2$*). For any two vectors $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ in $\mathbb{R}^2$, define

$$\begin{aligned} \langle \mathbf{x}, \mathbf{y} \rangle &= ax_1y_1 + c(x_1y_2 + x_2y_1) + bx_2y_2 \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & c \\ c & b \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{x}^T A \mathbf{y}, \end{aligned}$$

where a, b and c are arbitrary real numbers. Then, this function $\langle \cdot, \cdot \rangle$ clearly satisfies the first three rules of the inner product, i.e., $\langle \cdot, \cdot \rangle$ is symmetric and bilinear. Moreover, if $a > 0$ and $\det A = ab - c^2 > 0$ hold, then it also satisfies rule (4), the positive definiteness of the inner product. (Hint: The equation $\langle \mathbf{x}, \mathbf{x} \rangle = ax_1^2 + 2cx_1x_2 + bx_2^2 \geq 0$ if and only if either $x_2 = 0$ or the discriminant of $\langle \mathbf{x}, \mathbf{x} \rangle / x_2^2$ is nonpositive.)

In the case of $c = 0$, this reduces to $\langle \mathbf{x}, \mathbf{y} \rangle = ax_1y_1 + bx_2y_2$. Notice also that $a = \langle \mathbf{e}_1, \mathbf{e}_1 \rangle$, $b = \langle \mathbf{e}_2, \mathbf{e}_2 \rangle$ and $c = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \langle \mathbf{e}_2, \mathbf{e}_1 \rangle$. $\square$

Example 3.1.4 (*Case of $\mathbf{x} \neq \mathbf{0} \neq \mathbf{y}$ but $\langle \mathbf{x}, \mathbf{y} \rangle = 0$*). (1) Let $x = (1, 0)$, $y = (0, 1) \in \mathbb{R}^2$.

Then $x \neq 0 \neq y$, but $\langle x, y \rangle = 0$.

(2) Let $V = C[0, 1]$ be the vector space of all real-valued continuous functions on $[0, 1]$. For any two functions f and g in V , define

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx.$$

Then $\langle \cdot, \cdot \rangle$ is an inner product on V (verify this). Let

$$f(x) = \begin{cases} 1 - 2x & \text{if } 0 \leq x \leq \frac{1}{2}, \\ 0 & \text{if } \frac{1}{2} \leq x \leq 1, \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq \frac{1}{2}, \\ 2x - 1 & \text{if } \frac{1}{2} \leq x \leq 1. \end{cases}$$

Then $f \neq 0 \neq g$, but $\langle f, g \rangle = 0$.

By a **subspace** W of an inner product space V , we mean a subspace of the vector space V together with the inner product that is the restriction of the inner product on V to W , i.e., with the same inner product as that of V .

If W is a vector subspace of V , but has a different inner product than V , then W cannot be a subspace of the inner product space V . For instance, see the following example.

Example 3.1.5 (*A subspace as an inner product space*). The set $W = D^1[0, 1]$ of all real-valued *differentiable* functions on $[0, 1]$ is a subspace of $V = C[0, 1]$. The restriction to W of the inner product on V defined in Example 3.1.4 (2) above makes W an inner product subspace of V . However, one can define another inner product on W by the following formula: For any two functions $f(x)$ and $g(x)$ in W ,

$$\langle\langle f, g \rangle\rangle = \int_0^1 f(x)g(x)dx + \int_0^1 f'(x)g'(x)dx.$$

Then $\langle\langle \cdot, \cdot \rangle\rangle$ is also an inner product on W , which is different from the restriction to W of the inner product of V , and hence W with this new inner product is **not a subspace** of the inner product space V . $\square$

3.2 The lengths and angles of vectors

In this section, we study a geometry of an inner product space by introducing a length, an angle or a distance between two vectors. The following inequality will enable us to define **an angle** between two vectors in an **inner product space** V .

Theorem 3.2.1 (Cauchy-Schwarz inequality). *If $\mathbf{x}$ and $\mathbf{y}$ are vectors in an inner product space V , then*

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle.$$

Proof. If $\mathbf{x} = \mathbf{0}$, it is clear. Assume $\mathbf{x} \neq \mathbf{0}$. For any scalar t , we have

$$\begin{aligned} 0 &\leq \langle t\mathbf{x} + \mathbf{y}, t\mathbf{x} + \mathbf{y} \rangle = \langle t\mathbf{x}, t\mathbf{x} \rangle + \langle t\mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, t\mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle \\ &= \langle \mathbf{x}, \mathbf{x} \rangle t^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle t + \langle \mathbf{y}, \mathbf{y} \rangle \end{aligned}$$

This inequality implies that the polynomial $\langle \mathbf{x}, \mathbf{x} \rangle t^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle t + \langle \mathbf{y}, \mathbf{y} \rangle$ in t has either no real roots or a repeated real root. Therefore, its discriminant must be nonpositive:

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 - \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle \leq 0,$$

which implies the inequality. $\square$

Exercise 3.2.2. Prove that the equality in the Cauchy-Schwarz inequality holds if and only if the vectors $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

The lengths of vectors and angles between two vectors in an inner product space are defined in a similar way to the case of the Euclidean n -space.

Definition 3.2.3. Let V be an inner product space.

- (1) The **magnitude** $\|\mathbf{x}\|$ (or the **length**) of a vector $\mathbf{x}$ is defined by

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

- (2) The **distance** $d(\mathbf{x}, \mathbf{y})$ between two vectors $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|.$$

- (3) From the Cauchy-Schwarz inequality, we have $-1 \leq \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} \leq 1$ for any two nonzero vectors $\mathbf{x}$ and $\mathbf{y}$. Hence, there is a unique number $\theta \in [0, \pi]$ such that

$$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} \quad \text{or} \quad \langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta.$$

Such a number θ is called the **angle** between $\mathbf{x}$ and $\mathbf{y}$.

Example 3.2.4 (Infinitely many different inner products on $\mathbb{R}^2$ or $\mathbb{R}^3$). (1) In $\mathbb{R}^2$ equipped with an inner product $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 + 3x_2y_2$, the angle between $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (1, 0)$ is computed as

$$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{2}{\sqrt{5} \cdot 2} = 0.6324 \dots$$

Thus, $\theta = \cos^{-1}(\frac{2}{\sqrt{10}})$. Notice that in the Euclidean 2-space $\mathbb{R}^2$ with the dot product, the angle between $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (1, 0)$ is clearly $\frac{\pi}{4}$ and $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} = 0.7071 \dots$. It shows that the angle between two vectors depends actually on an inner product on a vector space.

(2) For any diagonal matrix $A = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$ with all $d_i > 0$,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T A \mathbf{y} = d_1 x_1 y_1 + d_2 x_2 y_2 + d_3 x_3 y_3$$

defines an inner product on $\mathbb{R}^3$. Thus, there are infinitely many different inner products on $\mathbb{R}^3$.

Exercise 3.2.5. Prove the following properties of length in an inner product space V : For any vectors $\mathbf{x}, \mathbf{y} \in V$,

- (1) $\|\mathbf{x}\| \geq 0$,
- (2) $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$,
- (3) $\|k\mathbf{x}\| = |k|\|\mathbf{x}\|$,
- (4) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (triangle inequality).

Exercise 3.2.6. Let V be an inner product space. Show that for any vectors $\mathbf{x}, \mathbf{y}$ and $\mathbf{z}$ in V ,

- (1) $d(\mathbf{x}, \mathbf{y}) \geq 0$,
- (2) $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$,
- (3) $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$,
- (4) $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$ (triangle inequality).

Definition 3.2.7. Two vectors $\mathbf{x}$ and $\mathbf{y}$ in an inner product space are said to be **orthogonal** (or **perpendicular**) if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

Lemma 3.2.8. Let V be an inner product space and let $\mathbf{x} \in V$. Then, the vector $\mathbf{x}$ is orthogonal to every vector $\mathbf{y}$ in V (i.e., $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ for all $\mathbf{y}$ in V) if and only if $\mathbf{x} = \mathbf{0}$.

Proof. If $\mathbf{x} = \mathbf{0}$, clearly $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{0}, \mathbf{y} \rangle = 0$ for all $\mathbf{y}$ in V .

Conversely, suppose that $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ for all $\mathbf{y}$ in V . Then, in particular, $\langle \mathbf{x}, \mathbf{x} \rangle = 0$, implying $\mathbf{x} = \mathbf{0}$ by positive definiteness. □

Corollary 3.2.9. Let V be an inner product space, and let $\alpha = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then, a vector $\mathbf{x}$ in V is orthogonal to every basis vector $\mathbf{v}_i$ in α if and only if $\mathbf{x} = \mathbf{0}$.

Proof. If $\langle \mathbf{x}, \mathbf{v}_i \rangle = 0$ for $i = 1, 2, \dots, n$, then for any $\mathbf{y} = y_1 \mathbf{v}_1 + \dots + y_n \mathbf{v}_n = \sum_{i=1}^n y_i \mathbf{v}_i \in V$, we have $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n y_i \langle \mathbf{x}, \mathbf{v}_i \rangle = 0$.

Thus, $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ for all $\mathbf{y} \in V$. Then by the above theorem, $\mathbf{x} = \mathbf{0}$. □

Example 3.2.10 (Pythagorean Theorem). Let V be an inner product space, and let $\mathbf{x}$ and $\mathbf{y}$ be any two nonzero vectors in V with angle θ . By bilinearity and symmetry of the inner product:

$$\begin{aligned}\|\mathbf{x} + \mathbf{y}\|^2 &= \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle \\ &= \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle.\end{aligned}$$

Substituting the geometric formula $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|\|\mathbf{y}\|\cos\theta$ gives the Law of Cosines:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\|\cos\theta.$$

Moreover, if $\mathbf{x}$ and $\mathbf{y}$ are orthogonal ($\mathbf{x} \perp \mathbf{y}$, meaning $\langle \mathbf{x}, \mathbf{y} \rangle = 0$), this deduces the **Pythagorean theorem**:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2.$$

The following theorem tells that a set of non-zero orthogonal vectors is linearly independent.

Theorem 3.2.11. *If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are nonzero mutually orthogonal vectors in an inner product space V (i.e., $\mathbf{x}_i \neq \mathbf{0}$ for each i , and $\langle \mathbf{x}_i, \mathbf{x}_j \rangle = 0$ for $i \neq j$), then they are linearly independent.*

Proof. Suppose that a linear combination of these vectors is equal to the zero vector $\mathbf{0}$:

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_k\mathbf{x}_k = \mathbf{0},$$

for some scalars $c_1, c_2, \dots, c_k \in \mathbb{R}$.

Taking the inner product of both sides with $\mathbf{x}_i$ for any fixed $i \in \{1, 2, \dots, k\}$,

$$\begin{aligned}0 &= \langle \mathbf{0}, \mathbf{x}_i \rangle = \langle c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_k\mathbf{x}_k, \mathbf{x}_i \rangle \\ &= c_1\langle \mathbf{x}_1, \mathbf{x}_i \rangle + \cdots + c_i\langle \mathbf{x}_i, \mathbf{x}_i \rangle + \cdots + c_k\langle \mathbf{x}_k, \mathbf{x}_i \rangle.\end{aligned}$$

Since the vectors are mutually orthogonal, $\langle \mathbf{x}_j, \mathbf{x}_i \rangle = 0$ for all $j \neq i$. Thus, the summation reduces to

$$0 = c_i\langle \mathbf{x}_i, \mathbf{x}_i \rangle = c_i\|\mathbf{x}_i\|^2.$$

Since each vector is nonzero ($\mathbf{x}_i \neq \mathbf{0}$), positive definiteness ensures $\|\mathbf{x}_i\| \neq 0$, which implies $\|\mathbf{x}_i\|^2 > 0$. Therefore, $c_i = 0$. Since this holds for every $i = 1, 2, \dots, k$, we have $c_1 = c_2 = \cdots = c_k = 0$. Consequently, the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are linearly independent. $\square$

Does the converse of the above theorem hold? (**Verify!**)

3.3 Matrix representations of inner products

Let an $n \times n$ diagonal matrix with positive diagonal entries A be given. Then one can show that $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T A \mathbf{y}$ defines an inner product on the n -space $\mathbb{R}^n$ as shown in Example 3.2.4 (2). Conversely, every inner product on a vector space can be expressed in such a matrix product

form. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space, and let $\alpha = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a fixed ordered basis for V . Then for any $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{v}_i$ and $\mathbf{y} = \sum_{j=1}^n y_j \mathbf{v}_j$ in V ,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n \sum_{j=1}^n x_i y_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle$$

holds. If we set $a_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$ for $i, j = 1, 2, \dots, n$, then these numbers constitute a symmetric matrix $A = [a_{ij}]$, since $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$. Thus, in matrix notation, the inner product may be written as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n \sum_{j=1}^n x_i y_j a_{ij} = [\mathbf{x}]_{\alpha}^T A [\mathbf{y}]_{\alpha}.$$

The matrix A is called the **matrix representation** of the inner product $\langle \cdot, \cdot \rangle$ with respect to the basis α .

Example 3.3.1 (*Matrix representation of an inner product*). (1) With respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ for the Euclidean n -space $\mathbb{R}^n$, the matrix representation of the dot product is the identity matrix, since $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$. Thus, for $\mathbf{x} = \sum_i x_i \mathbf{e}_i$ and $\mathbf{y} = \sum_j y_j \mathbf{e}_j \in \mathbb{R}^n$, the dot product is just the matrix product $\mathbf{x}^T \mathbf{y}$:

$$\mathbf{x} \cdot \mathbf{y} = [x_1 \ \dots \ x_n] \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = [x_1 \ \dots \ x_n] \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \mathbf{x}^T \mathbf{y}.$$

(2) On $V = P_2([0, 1])$, we define an inner product on V as

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx.$$

Then for a basis $\alpha = \{f_1(x) = 1, f_2(x) = x, f_3(x) = x^2\}$ for V , its matrix representation $A = [a_{ij}]$ is given as follows:

• **First row** ($i = 1$):

$$\begin{aligned} a_{11} &= \langle f_1, f_1 \rangle = \int_0^1 1 \cdot 1 \, dx = [x]_0^1 = 1, \\ a_{12} &= \langle f_1, f_2 \rangle = \int_0^1 1 \cdot x \, dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}, \\ a_{13} &= \langle f_1, f_3 \rangle = \int_0^1 1 \cdot x^2 \, dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}. \end{aligned}$$

• **Second row** ($i = 2$):

$$\begin{aligned} a_{21} &= a_{12} = \frac{1}{2}, \\ a_{22} &= \langle f_2, f_2 \rangle = \int_0^1 x \cdot x \, dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}, \\ a_{23} &= \langle f_2, f_3 \rangle = \int_0^1 x \cdot x^2 \, dx = \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{4}. \end{aligned}$$

- **Third row** ($i = 3$):

$$\begin{aligned} a_{31} &= a_{13} = \frac{1}{3}, \\ a_{32} &= a_{23} = \frac{1}{4}, \\ a_{33} &= \langle f_3, f_3 \rangle = \int_0^1 x^2 \cdot x^2 dx = \left[\frac{x^5}{5} \right]_0^1 = \frac{1}{5}. \end{aligned}$$

Therefore, the full matrix representation A of the inner product is

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}.$$

The expression of the dot product as a matrix product is very useful in stating or proving theorems in the Euclidean space.

For any symmetric matrix A and for a fixed basis α , the formula $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\alpha^T A [\mathbf{y}]_\alpha$ seems to give rise to an inner product on V . In fact, the formula clearly is symmetric and bilinear, but does not necessarily satisfy the fourth rule, *positive definiteness*. The following theorem gives a necessary condition for a symmetric matrix A to give rise to an inner product.

Theorem 3.3.2. *The matrix representation A of an inner product (with respect to any basis) on a vector space V is invertible. That is, $\det A \neq 0$.*

Proof. Let $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\alpha^T A [\mathbf{y}]_\alpha$ be an inner product in a vector space V with respect to a basis α , and let $A[\mathbf{y}]_\alpha = \mathbf{0}$ as a homogeneous system of linear equations. Then

$$\langle \mathbf{y}, \mathbf{y} \rangle = [\mathbf{y}]_\alpha^T A [\mathbf{y}]_\alpha = 0.$$

It implies that $A[\mathbf{y}]_\alpha = \mathbf{0}$ has only the trivial solution $\mathbf{y} = \mathbf{0}$, or equivalently A is invertible. $\square$

3.4 Gram-Schmidt orthogonalization

In an inner product space, any vector having length 1 is termed a **unit vector**. Whenever $\mathbf{x}$ is a nonzero vector, scaling it as $\frac{1}{\|\mathbf{x}\|} \mathbf{x}$ produces a unit vector. This scaling by the reciprocal of the norm is known as **normalization**.

Definition 3.4.1. A family of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ in an inner product space V is called **orthonormal** if

$$\langle \mathbf{x}_i, \mathbf{x}_j \rangle = \delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \quad (\text{orthogonality}), \\ 1 & \text{if } i = j \quad (\text{normality}). \end{cases}$$

An ordered set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is called an **orthonormal basis** for V if it constitutes a basis that is orthonormal.

Exercise 3.4.2. Establish whether each of the given pairs in $\mathbb{R}^2$ is orthogonal, orthonormal, or neither relative to the standard Euclidean inner product:

$$(1) \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix} \right\}$$

$$(3) \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\}$$

$$(2) \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \right\}$$

$$(4) \left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \right\}$$

Every finite-dimensional inner product space admits an orthonormal basis. The following example demonstrates the step-by-step construction of such a basis.

Example 3.4.3 (*Constructing an orthonormal basis*). Given the matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \\ 1 & 1 & 0 \end{bmatrix},$$

determine an orthonormal basis for the column space $\mathcal{C}(A)$.

Solution: Let $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ denote the successive column vectors of A . Because they are linearly independent, they form a basis for the 3-dimensional column space $\mathcal{C}(A) \subset \mathbb{R}^4$. Let $\text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ denote the subspace spanned by these elements.

(1) First normalize the leading vector $\mathbf{c}_1$:

$$\mathbf{u}_1 = \frac{\mathbf{c}_1}{\|\mathbf{c}_1\|} = \frac{\mathbf{c}_1}{2} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right),$$

which is a unit vector. Clearly, $\text{Span}\{\mathbf{u}_1\} = \text{Span}\{\mathbf{c}_1\}$.

(2) Next, the component $\mathbf{c}_2 - \langle \mathbf{u}_1, \mathbf{c}_2 \rangle \mathbf{u}_1 = \mathbf{c}_2 - 2\mathbf{u}_1 = (0, 1, -1, 0)$ is nonzero and orthogonal to $\mathbf{u}_1$. Normalizing it gives:

$$\mathbf{u}_2 = \frac{\mathbf{c}_2 - \langle \mathbf{u}_1, \mathbf{c}_2 \rangle \mathbf{u}_1}{\|\mathbf{c}_2 - \langle \mathbf{u}_1, \mathbf{c}_2 \rangle \mathbf{u}_1\|} = \frac{1}{\sqrt{2}}(0, 1, -1, 0) = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right).$$

Hence $\{\mathbf{u}_1, \mathbf{u}_2\}$ is orthonormal and satisfies $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\} = \text{Span}\{\mathbf{c}_1, \mathbf{c}_2\}$.

(3) Finally, observe that

$$\mathbf{c}_3 - \langle \mathbf{u}_1, \mathbf{c}_3 \rangle \mathbf{u}_1 - \langle \mathbf{u}_2, \mathbf{c}_3 \rangle \mathbf{u}_2 = \mathbf{c}_3 - 4\mathbf{u}_1 + \sqrt{2}\mathbf{u}_2 = (0, 1, 1, -2)$$

is orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$. Normalizing this vector yields:

$$\mathbf{u}_3 = \frac{\mathbf{c}_3 - \langle \mathbf{u}_1, \mathbf{c}_3 \rangle \mathbf{u}_1 - \langle \mathbf{u}_2, \mathbf{c}_3 \rangle \mathbf{u}_2}{\|\mathbf{c}_3 - \langle \mathbf{u}_1, \mathbf{c}_3 \rangle \mathbf{u}_1 - \langle \mathbf{u}_2, \mathbf{c}_3 \rangle \mathbf{u}_2\|} = \frac{1}{\sqrt{6}}(0, 1, 1, -2).$$

Thus, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ forms an orthonormal basis for $\mathcal{C}(A)$. □

Theorem 3.4.4 (*Gram-Schmidt Orthogonalization*). Every finite-dimensional inner product space possesses an orthonormal basis.

Proof. Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be any basis for an n -dimensional inner product space V . First we construct an **orthogonal** set of nonzero vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, and then we **normalize** each vector to obtain unit vectors $\mathbf{u}_k = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}$.

We prove by induction on k .

Step 1: Base Case ($k = 1$)

Choose the first direction directly, i.e., $\mathbf{v}_1 = \mathbf{x}_1$. Since $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is linearly independent, $\mathbf{x}_1 \neq \mathbf{0}$, which means $\|\mathbf{v}_1\| \neq 0$. Normalizing gives:

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\mathbf{x}_1}{\|\mathbf{x}_1\|}.$$

Clearly, $\|\mathbf{u}_1\| = 1$ and $\text{span}\{\mathbf{u}_1\} = \text{span}\{\mathbf{x}_1\}$.

Step 2: Induction Hypothesis

Assume that for some $k - 1$ (with $2 \leq k \leq n$), we have already constructed an orthonormal set $\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}\}$ such that $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}\} = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{k-1}\}$.

Step 3: Inductive Step (Constructing $\mathbf{u}_k$)

Let

$$\mathbf{v}_k = \mathbf{x}_k - \sum_{j=1}^{k-1} \langle \mathbf{u}_j, \mathbf{x}_k \rangle \mathbf{u}_j.$$

(1) **Orthogonality:** Take the inner product of $\mathbf{v}_k$ with any earlier vector $\mathbf{u}_i$ ($1 \leq i \leq k - 1$):

$$\begin{aligned} \langle \mathbf{u}_i, \mathbf{v}_k \rangle &= \left\langle \mathbf{u}_i, \mathbf{x}_k - \sum_{j=1}^{k-1} \langle \mathbf{u}_j, \mathbf{x}_k \rangle \mathbf{u}_j \right\rangle \\ &= \langle \mathbf{u}_i, \mathbf{x}_k \rangle - \sum_{j=1}^{k-1} \langle \mathbf{u}_j, \mathbf{x}_k \rangle \langle \mathbf{u}_i, \mathbf{u}_j \rangle. \\ &= \langle \mathbf{u}_i, \mathbf{x}_k \rangle - \langle \mathbf{u}_i, \mathbf{x}_k \rangle (1) \quad \left(\because \langle \mathbf{u}_i, \mathbf{u}_j \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases} \right) \\ &= 0. \end{aligned}$$

Hence, $\mathbf{v}_k$ is orthogonal to each of $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$.

(2) **Check that $\mathbf{v}_k \neq \mathbf{0}$:** If $\mathbf{v}_k = \mathbf{0}$, then

$$\mathbf{x}_k = \sum_{j=1}^{k-1} \langle \mathbf{u}_j, \mathbf{x}_k \rangle \mathbf{u}_j \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}\} = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{k-1}\},$$

which contradicts the assumption that $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is linearly independent. Thus, $\mathbf{v}_k \neq \mathbf{0}$, meaning $\|\mathbf{v}_k\| \neq 0$.

(3) **Normalize:** Define the unit vector: $\mathbf{u}_k = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}$. Then $\|\mathbf{u}_k\| = 1$.

(4) **Preserve the Span:** Clearly, by definition of $\mathbf{v}_k$,

$$\mathbf{u}_k = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}, \mathbf{x}_k\} = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{k-1}, \mathbf{x}_k\}.$$

Conversely,

$$\mathbf{x}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{u}_j, \mathbf{x}_k \rangle \mathbf{u}_j = \|\mathbf{v}_k\| \mathbf{u}_k + \sum_{j=1}^{k-1} \langle \mathbf{u}_j, \mathbf{x}_k \rangle \mathbf{u}_j \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}.$$

Therefore, $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Since any orthonormal set is linearly independent, this set of n vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is an orthonormal basis for n -dimensional space V . $\square$

Exercise 3.4.5. Use the Gram–Schmidt orthogonalization on the Euclidean space $\mathbb{R}^4$ to transform the basis

$$\{(0, 1, 1, 0), (-1, 1, 0, 0), (1, 2, 0, -1), (-1, 0, 0, -1)\}$$

into an orthonormal basis.

Exercise 3.4.6. Find an orthonormal basis for the subspace W of the Euclidean space $\mathbb{R}^3$ given by $x + 2y - z = 0$.

Exercise 3.4.7. Let $V = C[0, 1]$ with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx \quad \text{for any } f \text{ and } g \text{ in } V.$$

Find an orthonormal basis for the subspace spanned by $1, x$ and x^2 .

The next theorem shows that an orthonormal basis (for a finite-dimensional space) acts just like the standard basis for the Euclidean n -space $\mathbb{R}^n$.

Theorem 3.4.8. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be an orthonormal basis for a subspace U in an inner product space V . Then, for any vector $\mathbf{x}$ in U ,

$$\mathbf{x} = \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \langle \mathbf{u}_2, \mathbf{x} \rangle \mathbf{u}_2 + \dots + \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k.$$

Proof. For any vector $\mathbf{x} \in U$, one can write $\mathbf{x} = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \dots + x_k\mathbf{u}_k$, as a linear combination of the basis vectors. However, since $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is orthonormal, for each $i = 1, \dots, k$,

$$\begin{aligned} \langle \mathbf{u}_i, \mathbf{x} \rangle &= \langle \mathbf{u}_i, x_1\mathbf{u}_1 + \dots + x_k\mathbf{u}_k \rangle \\ &= x_1\langle \mathbf{u}_i, \mathbf{u}_1 \rangle + \dots + x_i\langle \mathbf{u}_i, \mathbf{u}_i \rangle + \dots + x_k\langle \mathbf{u}_i, \mathbf{u}_k \rangle \\ &= x_i. \end{aligned}$$

$\square$

Theorem 3.4.9. Any n -dimension inner product space V with an inner product $\langle \cdot, \cdot \rangle$ is isomorphic to the Euclidean n -space $\mathbb{R}^n$ with the dot product $\cdot$, which means that an isomorphism preserves the inner product.

Proof. Let $\alpha = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an orthonormal basis for the space V . With this orthonormal basis α , the natural isomorphism $\Phi : V \rightarrow \mathbb{R}^n$ is given by $\Phi(\mathbf{v}_i) = [\mathbf{v}_i]_\alpha = \mathbf{e}_i$, $i = 1, 2, \dots, n$, where $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is the standard basis for $\mathbb{R}^n$. We shall show that this natural isomorphism preserves the inner product on vectors.

For a vector $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{v}_i$ in V with $x_i = \langle \mathbf{x}, \mathbf{v}_i \rangle$, the coordinate vector of $\mathbf{x}$ with respect to α is a column matrix

$$\Phi(\mathbf{x}) = [\mathbf{x}]_{\alpha} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Moreover, for another vector $\mathbf{y} = \sum_{i=1}^n y_i \mathbf{v}_i$ in V ,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \left\langle \sum_{i=1}^n x_i \mathbf{v}_i, \sum_{j=1}^n y_j \mathbf{v}_j \right\rangle = \sum_{i=1}^n x_i y_i = [\mathbf{x}]_{\alpha}^T [\mathbf{y}]_{\alpha}.$$

The right-hand side of the above is just the dot product of vectors in $\mathbb{R}^n$. That is,

$$\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_{\alpha}^T [\mathbf{y}]_{\alpha} = \Phi(\mathbf{x}) \cdot \Phi(\mathbf{y}) = \langle \Phi(\mathbf{x}), \Phi(\mathbf{y}) \rangle$$

for any $\mathbf{x}, \mathbf{y} \in V$. Hence, the natural isomorphism Φ preserves the inner product. $\square$

3.5 Projections

Let U be a subspace of a vector space V . Then, by Corollary 3.13 there is another subspace W of V such that $V = U \oplus W$, so that any $\mathbf{x} \in V$ has a unique expression as $\mathbf{x} = \mathbf{u} + \mathbf{w}$ for $\mathbf{u} \in U$ and $\mathbf{w} \in W$. Let $T : V \rightarrow V$ be defined by $T(\mathbf{x}) = T(\mathbf{u} + \mathbf{w}) = \mathbf{u}$. Then it is an easy exercise (**verify!**) to show that T is linear with image $\text{Im}(T) = T(V) = U$ and kernel $\text{Ker}(T) = W$.

Definition 3.5.1. Let U and W be subspaces of a vector space V . A linear transformation $T : V \rightarrow V$ is called the **projection** of V onto the subspace U along W if $V = U \oplus W$ and $T(\mathbf{x}) = \mathbf{u}$ for $\mathbf{x} = \mathbf{u} + \mathbf{w} \in U \oplus W$.

Example 3.5.2 (*Infinitely many different projections of $\mathbb{R}^2$ onto the x -axis*). Let X, Y and Z be the 1-dimensional subspaces of the Euclidean 2-space $\mathbb{R}^2$ spanned by the vectors $\mathbf{e}_1, \mathbf{e}_2$, and $\mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2 = (1, 1)$, respectively, i.e.,

$$\begin{aligned} X &= \{r\mathbf{e}_1 : r \in \mathbb{R}\} = x\text{-axis}, \\ Y &= \{r\mathbf{e}_2 : r \in \mathbb{R}\} = y\text{-axis}, \\ Z &= \{r(\mathbf{e}_1 + \mathbf{e}_2) : r \in \mathbb{R}\}. \end{aligned}$$

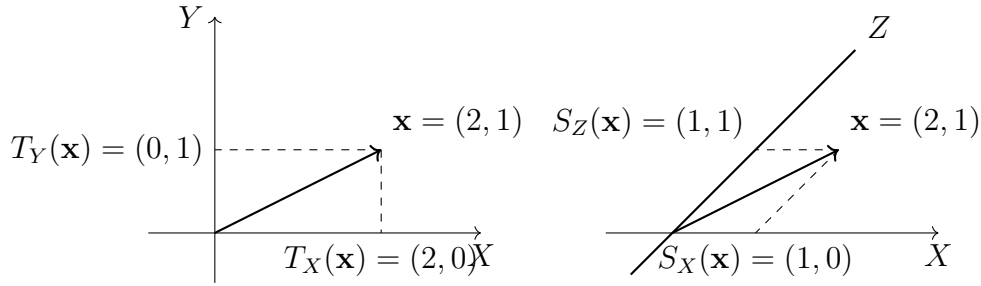
Since the pairs $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_1, \mathbf{v}\}$ are linearly independent, the space $\mathbb{R}^2$ can be expressed as the direct sum in two ways: $\mathbb{R}^2 = X \oplus Y = X \oplus Z$.

Thus, a vector $\mathbf{x} = (2, 1) \in \mathbb{R}^2$ may be written in two ways:

$$\mathbf{x} = (2, 1) = \begin{cases} 2(1, 0) + (0, 1) \in X \oplus Y = \mathbb{R}^2, & \text{or} \\ (1, 0) + (1, 1) \in X \oplus Z = \mathbb{R}^2. \end{cases}$$

Let T_X and S_X denote the projections of $\mathbb{R}^2$ onto X along Y and Z , respectively. Then

$$\begin{aligned} T_X(\mathbf{x}) &= 2(1, 0) \in X, & T_Y(\mathbf{x}) &= (0, 1) \in Y, & \text{and} \\ S_X(\mathbf{x}) &= (1, 0) \in X, & S_Z(\mathbf{x}) &= (1, 1) \in Z. \end{aligned}$$

Figure 3.1: Two decompositions of $\mathbb{R}^2$

This shows that a projection of $\mathbb{R}^2$ onto the subspace X depends on a choice of complementary subspace of X . For example, by choosing $Z_n = \{r(n, 1) : r \in \mathbb{R}\}$ for any integer n as a complementary subspace, one can construct infinitely many different projections of $\mathbb{R}^2$ onto the x -axis. $\square$

Theorem 3.5.3. A linear transformation $T : V \rightarrow V$ is a projection if and only if $T = T^2$, where $T^2 = T \circ T$ by definition.

Corollary 3.5.4. A linear transformation $T : V \rightarrow V$ is a projection if and only if $\text{id}_V - T$ is a projection. Moreover, if T is the projection of V onto a subspace U along W , then $\text{id}_V - T$ is the projection of V onto W along U .

Proof. It is enough to show that $(\text{id}_V - T) \circ (\text{id}_V - T) = \text{id}_V - T$. But

$$(\text{id}_V - T) \circ (\text{id}_V - T) = (\text{id}_V - T) - (T - T^2) = \text{id}_V - T.$$

 $\square$

3.5.1 Orthogonal projections

Definition 3.5.5. Let U and W be subspaces of an inner product space V .

- (1) Two subspaces U and W are said to be **orthogonal**, written $U \perp W$, if $\langle \mathbf{u}, \mathbf{w} \rangle = 0$ for each $\mathbf{u} \in U$ and $\mathbf{w} \in W$.
- (2) The set of all vectors in V that are orthogonal to every vector in U is called the **orthogonal complement** of U , denoted by $U^\perp$, i.e.,

$$U^\perp = \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

One can easily show that $U^\perp$ is a subspace of V , and $\mathbf{v} \in U^\perp$ if and only if $\langle \mathbf{v}, \mathbf{u} \rangle = 0$ for every $\mathbf{u} \in \beta$, where β is a basis for U . Moreover, $W \perp U$ if and only if $W \subseteq U^\perp$.

Exercise 3.5.6. Let U and W be subspaces of an inner product space V . Show that

- (1) If $U \perp W$, $U \cap W = \{\mathbf{0}\}$.
- (2) $U \subseteq W$ if and only if $W^\perp \subseteq U^\perp$.

Theorem 3.5.7. Let U be a subspace of an inner product space V . Then

- (1) $(U^\perp)^\perp = U$.
- (2) $V = U \oplus U^\perp$; that is, for each $\mathbf{x} \in V$, there exist unique vectors $\mathbf{x}_U \in U$ and $\mathbf{x}_{U^\perp} \in U^\perp$ such that $\mathbf{x} = \mathbf{x}_U + \mathbf{x}_{U^\perp}$. This is called the **orthogonal decomposition** of V (or of $\mathbf{x}$) by U .

Proof. Let $\dim U = k$. To show $(U^\perp)^\perp = U$, take an orthonormal basis for U , say $\alpha = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, by the Gram–Schmidt orthogonalization, and then extend it to an orthonormal basis for V , say $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n\}$, which is always possible (as in Corollary 1.4.11). Then, clearly $\gamma = \{\mathbf{v}_{k+1}, \dots, \mathbf{v}_n\}$ forms an (orthonormal) basis for $U^\perp$ (**check!**), which means that $(U^\perp)^\perp = U$ and $V = U \oplus U^\perp$. $\square$

Definition 3.5.8. Let U be a subspace of an inner product space V , and let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ be an orthonormal basis for U . The **orthogonal projection** Proj_U from V onto the subspace U is defined by

$$\text{Proj}_U(\mathbf{x}) = \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \langle \mathbf{u}_2, \mathbf{x} \rangle \mathbf{u}_2 + \cdots + \langle \mathbf{u}_m, \mathbf{x} \rangle \mathbf{u}_m$$

for any $\mathbf{x} \in V$.

Almost all projections used in linear algebra are orthogonal projections.

Example 3.5.9 (*The orthogonal projection from $\mathbb{R}^3$ onto the xy -plane*). In the Euclidean 3-space $\mathbb{R}^3$, let U be the xy -plane with the orthonormal basis $\alpha = \{\mathbf{e}_1, \mathbf{e}_2\}$. Then, the orthogonal projection $\text{Proj}_U(\mathbf{x}) = \langle \mathbf{e}_1, \mathbf{x} \rangle \mathbf{e}_1 + \langle \mathbf{e}_2, \mathbf{x} \rangle \mathbf{e}_2$ is the orthogonal projection onto the xy -plane in a usual sense in geometry, and $\mathbf{x} - \text{Proj}_U(\mathbf{x}) \in U^\perp$, which is the z -axis. It actually means that $\text{Proj}_U(x_1, x_2, x_3) = (x_1, x_2, 0)$ for any $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$. $\square$

Example 3.5.10 (*The orthogonal projection from $\mathbb{R}^2$ onto the x -axis*). As in Example 5.8, let X , Y and Z be the 1-dimensional subspaces of the Euclidean 2-space $\mathbb{R}^2$ spanned by the vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2 = (1, 1)$, respectively. Then clearly $Y = X^\perp$ and $V \neq X^\perp$. And, for the projections T_X and S_X of $\mathbb{R}^2$ given in Example 3.5.2, T_X is the orthogonal projection, but S_X is not, so that $T_X = \text{Proj}_X$ and $S_X \neq \text{Proj}_X$. $\square$

3.6 Diagonalization

3.6.1 Eigenvalues and eigenvectors

Definition 3.6.1. Let A be an $n \times n$ square matrix. A nonzero vector $\mathbf{x}$ in the n -space $\mathbb{R}^n$ is called an **eigenvector** (or **characteristic vector**) of A if there is a scalar λ in $\mathbb{R}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}.$$

The scalar λ is called an **eigenvalue** (or **characteristic value**) of A , and we say $\mathbf{x}$ **belongs** to λ .

Geometrically, an eigenvector of a matrix A is a nonzero vector $\mathbf{x}$ in the n -space $\mathbb{R}^n$ such that the vectors $\mathbf{x}$ and $A\mathbf{x}$ are parallel. In other words, the subspace W spanned by $\mathbf{x}$ is *invariant* under the linear transformation $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ in the sense $A(W) \subset W$. Algebraically, an eigenvector $\mathbf{x}$ is a nontrivial solution of the homogeneous system $(\lambda I - A)\mathbf{x} = \mathbf{0}$ of linear equations, that is, an eigenvector $\mathbf{x}$ is a nonzero vector in the null space $\mathcal{N}(\lambda I - A)$.

There are two unknowns in the system $(\lambda I - A)\mathbf{x} = \mathbf{0}$: an eigenvalue λ and an eigenvector $\mathbf{x}$. To find those unknowns, first we should determine an eigenvalue λ by using the fact that the equation $(\lambda I - A)\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\mathbf{x}$ if and only if λ satisfies the equation

$$\det(\lambda I - A) = 0,$$

called the **characteristic equation** of A . Note that $\det(\lambda I - A)$ is a polynomial of degree n in λ and it will be called the **characteristic polynomial** of A . Thus, the eigenvalues are just the roots of the characteristic equation $\det(\lambda I - A) = 0$.

Next, the eigenvectors of A can be determined by solving the homogeneous system $(\lambda I - A)\mathbf{x} = \mathbf{0}$ for each eigenvalue λ .

Recall that the eigenvectors of A belonging to an eigenvalue λ are just the nonzero vectors $\mathbf{x}$ in the null space $\mathcal{N}(\lambda I - A)$. This null space is called the **eigenspace** of A belonging to λ , and denoted by $E(\lambda)$.

Example 3.6.2 (*Matrix having distinct eigenvalues*). Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix}.$$

Solution: The characteristic polynomial is

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 2 & -\sqrt{2} \\ -\sqrt{2} & \lambda - 1 \end{bmatrix} = \lambda^2 - 3\lambda = \lambda(\lambda - 3).$$

Thus the eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = 3$. To determine the eigenvectors belonging to λ_i 's, we should solve the homogeneous system of equations $(\lambda_i I - A)\mathbf{x} = \mathbf{0}$. Let us take $\lambda_1 = 0$ first; then the system of equations $(\lambda_1 I - A)\mathbf{x} = \mathbf{0}$ becomes

$$\begin{cases} -2x_1 - \sqrt{2}x_2 = 0, \\ -\sqrt{2}x_1 - x_2 = 0, \end{cases} \quad \text{or} \quad x_2 = -\sqrt{2}x_1.$$

Hence, $\mathbf{x}_1 = (x_1, x_2) = (-1, \sqrt{2})$ is an eigenvector belonging to $\lambda_1 = 0$, and

$$E(0) = \{t\mathbf{x}_1 : t \in \mathbb{R}\}.$$

Note that one can take any nonzero solution (x_1, x_2) as an eigenvector $\mathbf{x}_1$ belonging to $\lambda_1 = 0$.

For $\lambda_2 = 3$, the system of equations $(\lambda_2 I - A)\mathbf{x} = \mathbf{0}$ becomes

$$\begin{cases} x_1 - \sqrt{2}x_2 = 0, \\ -\sqrt{2}x_1 + 2x_2 = 0, \end{cases} \quad \text{or} \quad x_1 = \sqrt{2}x_2.$$

Thus, taking $x_2 = 1$, we get $\mathbf{x}_2 = (\sqrt{2}, 1)$ as one of the eigenvectors belonging to $\lambda_2 = 3$ and so

$$E(3) = \{t\mathbf{x}_2 : t \in \mathbb{R}\}.$$

Note that the eigenvectors $\mathbf{x}_1$ and $\mathbf{x}_2$ belonging to the eigenvalues λ_1 and λ_2 respectively are linearly independent. $\square$

Example 3.6.3 (*Matrix having a repeated eigenvalue but full eigenvectors*). Find a basis for the eigenspaces of

$$A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Solution: The characteristic polynomial of A is $(\lambda - 1)(\lambda - 5)^2$, so that the eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = 5$ with multiplicity 2. Thus, there are two eigenspaces of A . By definition, $\mathbf{x} = (x_1, x_2, x_3)$ is an eigenvector of A belonging to λ if and only if $\mathbf{x}$ is a nontrivial solution of the homogeneous system $(\lambda I - A)\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} \lambda - 3 & 2 & 0 \\ 2 & \lambda - 3 & 0 \\ 0 & 0 & \lambda - 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

If $\lambda_1 = 1$, then the system becomes

$$\begin{bmatrix} -2 & 2 & 0 \\ 2 & -2 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Solving this system yields $x_1 = t$, $x_2 = t$, $x_3 = 0$ for $t \in \mathbb{R}$. Thus, the eigenvectors belonging to $\lambda_1 = 1$ are nonzero vectors of the form

$$\mathbf{x} = \begin{bmatrix} t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad t \in \mathbb{R},$$

so that $(1, 1, 0)$ is a basis for the eigenspace $E(\lambda_1)$ belonging to $\lambda_1 = 1$.

If $\lambda_2 = 5$, then the system becomes

$$\begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Solving this system yields $x_1 = -s$, $x_2 = s$, $x_3 = t$ for $s, t \in \mathbb{R}$. Thus, the eigenvectors of A belonging to $\lambda_2 = 5$ are nonzero vectors of the form

$$\mathbf{x} = \begin{bmatrix} -s \\ s \\ t \end{bmatrix} = \begin{bmatrix} -s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

for $s, t \in \mathbb{R}$. Since $(-1, 1, 0)$ and $(0, 0, 1)$ are linearly independent, they form a basis for the eigenspace $E(\lambda_2)$ belonging to $\lambda_2 = 5$. $\square$

For each eigenvalue λ of A in the above two examples, one can see that the dimension of the eigenspace $E(\lambda)$ is equal to the multiplicity of λ as a root of the equation $\det(\lambda I - A) = 0$. But, in general it is **not true** as the next example shows.

Example 3.6.4 (*Matrix having a repeated eigenvalue with insufficient eigenvectors*). Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}.$$

A simple computation shows that the characteristic polynomial of the matrix A is $(\lambda - 2)^5$ so that the eigenvalue $\lambda = 2$ is of multiplicity 5. However, there is only one linearly independent eigenvector $\mathbf{e}_1 = (1, 0, 0, 0, 0)$ belonging to $\lambda = 2$ because $\text{rank}(2I - A) = 4$, which shows that $\dim E(\lambda) = \dim \mathcal{N}(2I - A) = 1$ is less than the multiplicity of λ . $\square$

Note that the equation $\det(\lambda I - A) = 0$ may have complex roots, which are called **complex eigenvalues**. However, the complex numbers are not scalars of the real vector space. In many cases, it is necessary to deal with those complex numbers, that is, we need to expand the set of scalars to the set of complex numbers. This expansion of the set of scalars to the set of complex numbers leads us to work with complex vector spaces. In this unit and course, we restrict our discussion to the case of real eigenvalues.

Example 3.6.5 (*Matrix having complex eigenvalues*). The characteristic polynomial of the matrix

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

is $\lambda^2 - 2 \cos \theta \lambda + (\cos^2 \theta + \sin^2 \theta)$. Thus, the eigenvalues are $\lambda = \cos \theta \pm i \sin \theta$ (obtained by finding roots of quadratic equation using $b^2 - 4ac$), which are complex numbers, so this matrix as a rotation of $\mathbb{R}^2$ has no real eigenvalues unless $\theta = n\pi$, $n = 0, \pm 1, \pm 2, \dots$ $\square$

In the following, we derive some basic properties of the eigenvalues and eigenvectors.

- Lemma 3.6.6.** (1) *If A is a triangular matrix, then the diagonal entries are exactly the eigenvalues of A .*
 (2) *If A and B are square matrices similar to each other, then they have the same characteristic polynomial.*

Proof. (1) The characteristic equation of an upper triangular matrix is

$$\begin{aligned} \det(\lambda I - A) &= \det \begin{bmatrix} \lambda - a_{11} & * & \dots & * \\ & \lambda - a_{22} & \dots & * \\ & & \ddots & \vdots \\ \mathbf{0} & & & \lambda - a_{nn} \end{bmatrix} \\ &= (\lambda - a_{11}) \dots (\lambda - a_{nn}) = 0. \end{aligned}$$

(2) Since there exists a nonsingular matrix Q such that $B = Q^{-1}AQ$,

$$\begin{aligned} \det(\lambda I - B) &= \det(Q^{-1}(\lambda I)Q - Q^{-1}AQ) \\ &= \det(Q^{-1}(\lambda I - A)Q) \\ &= \det Q^{-1} \det(\lambda I - A) \det Q \\ &= \det(\lambda I - A). \end{aligned}$$

□

Part (2) in the above lemma says that similar matrices have the same eigenvalues, i.e., *the eigenvalues are invariant under the similarity*. However, their eigenvectors might be different: in fact, $\mathbf{x}$ is an eigenvector of B belonging to λ if and only if $Q\mathbf{x}$ is an eigenvector of A belonging to λ , since $AQ = QB$, and

$$AQ\mathbf{x} = QB\mathbf{x} = \lambda Q\mathbf{x}.$$

Theorem 3.6.7. *Let an $n \times n$ matrix A have n eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ possibly with repetition. Then,*

- (1) $\det A = \lambda_1 \lambda_2 \dots \lambda_n$, (the product of the n eigenvalues),
 (2) $\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$, (the sum of the n eigenvalues).

Proof. (1) Since eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ are the zeros of the characteristic polynomial of A , we have

$$\det(\lambda I - A) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n).$$

If we take $\lambda = 0$ in both sides, then we get

$$(-1)^n \det A = (-1)^n \lambda_1 \lambda_2 \dots \lambda_n.$$

(2) On the other hand,

$$\begin{aligned} (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n) &= \det(\lambda I - A) \\ &= \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \dots & \lambda - a_{nn} \end{bmatrix}, \end{aligned}$$

which is a polynomial of the form $p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0$ in λ . Computing the coefficient c_{n-1} of λ^{n-1} in two ways by expanding both sides, we get

$$\lambda_1 + \lambda_2 + \dots + \lambda_n = a_{11} + a_{22} + \dots + a_{nn} = \operatorname{tr}(A).$$

□

Exercise 3.6.8. Show that:

- (1) for any 2×2 matrix A , $\det(\lambda I - A) = \lambda^2 - \operatorname{tr}(A)\lambda + \det A$;

In the above theorem, we assume that the matrix A has n (real) eigenvalues counting multiplicities. But, by allowing the scalars to be complex numbers, every $n \times n$ matrix has n eigenvalues counting multiplicities, so that the above theorem remains true for any square matrix.

Corollary 3.6.9. *The determinant and the trace of A are invariant under similarity.*

Recall that any square matrix A is singular if and only if $\det A = 0$. However, $\det A$ is the product of its n eigenvalues. Thus a square matrix A is singular if and only if zero is an eigenvalue of A , or A is invertible if and only if zero is not an eigenvalue of A .

The following corollaries are easy consequences of this fact.

Corollary 3.6.10. *For any $n \times n$ matrices A and B , the following are equivalent:*

- (1) *Zero is an eigenvalue of AB .*
- (2) *A or B is singular.*
- (3) *Zero is an eigenvalue of BA .*

Proof. Exercise. □

Corollary 3.6.11. *For any $n \times n$ matrices A and B , the matrices AB and BA have the same eigenvalues.*

Proof. By the above corollary, zero is an eigenvalue of AB if and only if it is an eigenvalue of BA . Let λ be a nonzero eigenvalue of AB with $(AB)\mathbf{x} = \lambda\mathbf{x}$ for a nonzero vector $\mathbf{x}$. Then the vector $B\mathbf{x}$ is not zero, since $\lambda \neq 0$, but

$$(BA)(B\mathbf{x}) = B(AB\mathbf{x}) = B(\lambda\mathbf{x}) = \lambda(B\mathbf{x}).$$

This means that $B\mathbf{x}$ is an eigenvector of BA belonging to the eigenvalue λ , and λ is an eigenvalue of BA . Similarly, any nonzero eigenvalue of BA is also an eigenvalue of AB . □

Exercise 3.6.12. Find the matrices A and B such that $\det A = \det B$, $\operatorname{tr}(A) = \operatorname{tr}(B)$, but A is not similar to B .

Exercise 3.6.13. Show that A and A^T have the same eigenvalues. Do they necessarily have the same eigenvectors?

Exercise 3.6.14. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of an $n \times n$ matrix A . Then

- (1) A is invertible if and only if $\lambda_i \neq 0$ for all $i = 1, 2, \dots, n$.
- (2) If A is invertible, then the inverse A^{-1} has eigenvalues $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$.

3.7 Diagonalization of matrices

Definition 3.7.1. A square matrix A is said to be **diagonalizable** if there exists an invertible matrix Q such that $Q^{-1}AQ$ is a diagonal matrix (i.e., A is similar to a diagonal matrix).

The next theorem characterizes a diagonalizable matrix, and the proof shows a practical way of diagonalizing a matrix.

Theorem 3.7.2. *Let A be an $n \times n$ matrix. Then A is diagonalizable if and only if A has n linearly independent eigenvectors.*

Proof. ($\implies$) Suppose A is diagonalizable. Then there is an invertible matrix Q such that $Q^{-1}AQ$ is a diagonal matrix D , say

$$Q^{-1}AQ = D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix},$$

or, equivalently, $AQ = QD$. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ denote the column vectors of Q . Since

$$\begin{aligned} AQ &= [A\mathbf{x}_1 \ A\mathbf{x}_2 \ \dots \ A\mathbf{x}_n], \\ QD &= [\lambda_1\mathbf{x}_1 \ \lambda_2\mathbf{x}_2 \ \dots \ \lambda_n\mathbf{x}_n], \end{aligned}$$

the matrix equation $AQ = QD$ implies $A\mathbf{x}_i = \lambda_i\mathbf{x}_i$ for $i = 1, \dots, n$. Moreover, since Q is invertible, their column vectors are nonzero and are linearly independent, that is, the $\mathbf{x}_i$'s are n linearly independent eigenvectors of A .

($\impliedby$) Assume that A has n linearly independent eigenvectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ belonging to the eigenvalues $\lambda_1, \dots, \lambda_n$, respectively, so that $A\mathbf{x}_i = \lambda_i\mathbf{x}_i$ for $i = 1, \dots, n$. If we define a matrix Q as

$$Q = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_n]$$

with $\mathbf{x}_j$ as the j -th column vector, then the same equation shows $AQ = QD$, where D is the diagonal matrix having the eigenvalues $\lambda_1, \dots, \lambda_n$ on the diagonal. Since the column vectors of Q are assumed to be linearly independent, Q is invertible, so $Q^{-1}AQ = D$. $\square$

Remark 3.7.3. (1) The proof of the above theorem reveals how to diagonalize an $n \times n$ matrix A .

Step 1 Find n linearly independent eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ of A .

Step 2 Form the matrix $Q = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_n]$.

Step 3 The matrix $Q^{-1}AQ$ will be a diagonal matrix with $\lambda_1, \dots, \lambda_n$ as its successive diagonal entries, where λ_j is the eigenvalue associated with the eigenvector $\mathbf{x}_j$, $j = 1, 2, \dots, n$.

- (2) Let α denote the standard basis for $\mathbb{R}^n$ and let $\beta = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be the basis for $\mathbb{R}^n$ consisting of n linearly independent eigenvectors of A . Then the matrix

$$Q = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_n] = [[\mathbf{x}_1]_\alpha \ [\mathbf{x}_2]_\alpha \ \dots \ [\mathbf{x}_n]_\alpha] = [id]_\beta^\alpha$$

is the basis-change matrix from β to α , and the matrix representation of A , as a linear transformation, with respect to β , is

$$[A]_\beta = [id]_\alpha^\beta [A]_\alpha [id]_\beta^\alpha = Q^{-1}AQ = \begin{bmatrix} \lambda_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \lambda_n \end{bmatrix}.$$

Note that the diagonal entries λ_i 's are the eigenvalues of A .

(3) Not all matrices are diagonalizable. A standard example is $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Its eigenvalues are $\lambda_1 = \lambda_2 = 0$. Hence, if A is diagonalizable, then

$$Q^{-1}AQ = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \mathbf{0}$$

for some invertible matrix Q , and then A must be the zero matrix. Since A is not the zero matrix, no invertible matrix Q can be obtained so that $Q^{-1}AQ$ is diagonal.

Example 3.7.4 (Several different types of a diagonalization). Diagonalize the matrix

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{bmatrix}.$$

Solution: A direct calculation gives that the eigenvalues of A are $\lambda_1 = 1$, $\lambda_2 = 1$ and $\lambda_3 = -2$, and their associated eigenvectors are

$$\mathbf{x}_1 = (1, 0, 0), \quad \mathbf{x}_2 = (0, 1, 1) \quad \text{and} \quad \mathbf{x}_3 = (1, 2, 1),$$

respectively. They are linearly independent, and the first two vectors $\mathbf{x}_1, \mathbf{x}_2$ form a basis for the eigenspace $E(1)$ belonging to $\lambda_1 = \lambda_2 = 1$, and $\mathbf{x}_3$ forms a basis for the eigenspace $E(-2)$ belonging to $\lambda_3 = -2$. Thus, the matrix

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

diagonalizes A . In fact, one can verify that

$$\begin{aligned} P^{-1}AP &= \begin{bmatrix} 1 & -1 & 1 \\ 0 & -1 & 2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 3 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \end{aligned}$$

Note that nothing would happen if we chose different eigenvectors belonging to the eigenvalues. For example, $\{(-1, 0, 0), (0, -1, -1)\}$ is another basis for $E(1)$, and $\{(2, 4, 2)\}$ is also a basis for $E(-2)$. The matrix

$$Q = \begin{bmatrix} -1 & 0 & 2 \\ 0 & -1 & 4 \\ 0 & -1 & 2 \end{bmatrix} \quad \text{also diagonalizes } A \text{ as } Q^{-1}AQ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \square$$

Exercise 3.7.5. Show that the following matrices are not diagonalizable.

$$(1) \quad A = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix},$$

$$(2) \ B = \begin{bmatrix} \lambda & 0 & 0 \\ 1 & \lambda & 0 \\ 0 & 1 & \lambda \end{bmatrix}, \quad \lambda \text{ is any scalar.}$$

Exercise 3.7.6. Construct a 2×2 matrix A whose eigenvalues are 2 and 3, and whose eigenvectors are $(2, 1)$ and $(3, 2)$, respectively.

The following theorem says that eigenvectors corresponding to distinct eigenvalues are linearly independent.

Theorem 3.7.7. *Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of a matrix A and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ eigenvectors belonging to them, respectively. Then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is linearly independent.*

Proof. Let r be the largest integer such that $\{\mathbf{x}_1, \dots, \mathbf{x}_r\}$ is linearly independent. If $r = k$, then there is nothing to prove. Suppose not, i.e., $1 \leq r < k$. Then $\square$

$\{\mathbf{x}_1, \dots, \mathbf{x}_{r+1}\}$ is linearly dependent. Thus, there exist scalars $c_1, c_2, \dots, c_{r+1}$ with $c_{r+1} \neq 0$ such that

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_{r+1}\mathbf{x}_{r+1} = \mathbf{0}. \quad (3.1)$$

Multiplying both sides by A and using

$$A\mathbf{x}_1 = \lambda_1\mathbf{x}_1, \ A\mathbf{x}_2 = \lambda_2\mathbf{x}_2, \ \dots, \ A\mathbf{x}_{r+1} = \lambda_{r+1}\mathbf{x}_{r+1},$$

one can get

$$c_1\lambda_1\mathbf{x}_1 + c_2\lambda_2\mathbf{x}_2 + \dots + c_{r+1}\lambda_{r+1}\mathbf{x}_{r+1} = \mathbf{0}. \quad (3.2)$$

Multiplying both sides of (3.1) by λ_{r+1} and subtracting the resulting equation from (3.2) yields

$$c_1(\lambda_1 - \lambda_{r+1})\mathbf{x}_1 + c_2(\lambda_2 - \lambda_{r+1})\mathbf{x}_2 + \dots + c_r(\lambda_r - \lambda_{r+1})\mathbf{x}_r = \mathbf{0}.$$

Since $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r\}$ is linearly independent and $\lambda_1, \lambda_2, \dots, \lambda_{r+1}$ are all distinct, it follows that $c_1 = c_2 = \dots = c_r = 0$. Substituting these values in (3.1) yields $c_{r+1} = 0$, which is a contradiction to the assumption. $\square$

As a consequence of the last two theorems, we have the following result.

Theorem 3.7.8. *If an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.*

The next example shows a simple application of the diagonalization to the computation of the power A^n of a matrix A .

Example 3.7.9. Compute A^{100} for the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}.$$

Solution:

Step 1: Find the eigenvalues of A .

The characteristic polynomial of A is obtained by setting $\det(\lambda I - A) = 0$:

$$\begin{aligned}\det(\lambda I - A) &= \det \begin{bmatrix} \lambda - 1 & -4 \\ -3 & \lambda - 2 \end{bmatrix} \\ &= (\lambda - 1)(\lambda - 2) - (-4)(-3) \\ &= \lambda^2 - 3\lambda - 10 \\ &= (\lambda - 5)(\lambda + 2) = 0.\end{aligned}$$

Thus, the eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = -2$.

Step 2: Find the corresponding eigenvectors.

To find the eigenvectors, we solve the homogeneous linear system $(\lambda I - A)\mathbf{x} = \mathbf{0}$ for each eigenvalue:

- **For $\lambda_1 = 5$:**

$$(5I - A)\mathbf{x} = \begin{bmatrix} 4 & -4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x_1 - x_2 = 0 \implies x_1 = x_2.$$

Choosing $x_2 = 1$ gives the eigenvector:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- **For $\lambda_2 = -2$:**

$$(-2I - A)\mathbf{x} = \begin{bmatrix} -3 & -4 \\ -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 3x_1 + 4x_2 = 0 \implies x_1 = -\frac{4}{3}x_2.$$

Choosing $x_2 = 3$ yields an integer eigenvector:

$$\mathbf{x}_2 = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$$

Step 3: Construct the diagonalizing matrix Q and compute Q^{-1} .

Form the matrix Q by setting the eigenvectors as column vectors:

$$Q = [\mathbf{x}_1 \ \mathbf{x}_2] = \begin{bmatrix} 1 & -4 \\ 1 & 3 \end{bmatrix}.$$

The corresponding diagonal matrix is

$$D = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix}.$$

The determinant of Q is $\det(Q) = (1)(3) - (-4)(1) = 7$. By the 2×2 inverse formula,

$$Q^{-1} = \frac{1}{7} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix}.$$

Step 4: Compute A^{100} using diagonalization.

Since $A = QDQ^{-1}$, taking the 100-th power yields

$$A^{100} = (QDQ^{-1})^{100} = QD^{100}Q^{-1}.$$

Since $(-2)^{100} = 2^{100}$, the diagonal power matrix is

$$D^{100} = \begin{bmatrix} 5^{100} & 0 \\ 0 & (-2)^{100} \end{bmatrix} = \begin{bmatrix} 5^{100} & 0 \\ 0 & 2^{100} \end{bmatrix}.$$

Now, carry out the matrix multiplication step-by-step:

(1) First multiply Q and D^{100} :

$$QD^{100} = \begin{bmatrix} 1 & -4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 5^{100} & 0 \\ 0 & 2^{100} \end{bmatrix} = \begin{bmatrix} 5^{100} & -4 \cdot 2^{100} \\ 5^{100} & 3 \cdot 2^{100} \end{bmatrix}.$$

(2) Next multiply by Q^{-1} :

$$\begin{aligned} A^{100} &= (QD^{100})Q^{-1} \\ &= \begin{bmatrix} 5^{100} & -4 \cdot 2^{100} \\ 5^{100} & 3 \cdot 2^{100} \end{bmatrix} \cdot \frac{1}{7} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix} \\ &= \frac{1}{7} \begin{bmatrix} (5^{100})(3) + (-4 \cdot 2^{100})(-1) & (5^{100})(4) + (-4 \cdot 2^{100})(1) \\ (5^{100})(3) + (3 \cdot 2^{100})(-1) & (5^{100})(4) + (3 \cdot 2^{100})(1) \end{bmatrix} \\ &= \frac{1}{7} \begin{bmatrix} 3 \cdot 5^{100} + 4 \cdot 2^{100} & 4 \cdot 5^{100} - 4 \cdot 2^{100} \\ 3 \cdot 5^{100} - 3 \cdot 2^{100} & 4 \cdot 5^{100} + 3 \cdot 2^{100} \end{bmatrix}. \end{aligned}$$

□

Exercise 3.7.10. For the matrix $A = \begin{bmatrix} 5 & -4 & 4 \\ 12 & -11 & 12 \\ 4 & -4 & 5 \end{bmatrix}$,

- (1) diagonalize the matrix A ; and
- (2) find the eigenvalues of $A^{10} + A^7 + 5A$.

3.8 Exponential matrices

Just like the Maclaurin series of the exponential function e^x , we define the exponential of a matrix.

Definition 3.8.1. For any square matrix A , the **exponential matrix** of A is defined as the series

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

That is, the exponential matrix e^A is defined to be the (entry-wise) limit of the sequence:

$$[e^A]_{ij} = \lim_{m \rightarrow \infty} \left[\sum_{k=0}^m \frac{A^k}{k!} \right]_{ij} \text{ for all } i, j.$$

Example 3.8.2. If

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad \text{then} \quad D^k = \begin{bmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{bmatrix} \quad \text{for any } k \geq 0.$$

Thus, the exponential matrix e^D is

$$e^D = \sum_{k=0}^{\infty} \frac{D^k}{k!} = \begin{bmatrix} \sum_{k=0}^{\infty} \frac{\lambda_1^k}{k!} & & 0 \\ & \ddots & \\ 0 & & \sum_{k=0}^{\infty} \frac{\lambda_n^k}{k!} \end{bmatrix} = \begin{bmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{bmatrix},$$

which coincides with the definition given on page 230. $\square$

Theorem 3.8.3. *For any square matrix A , the matrix e^A exists. In other words, each (i, j) -entry of e^A is convergent.*

Proof. Since A has only n^2 entries, there is a number M such that $|a_{ij}| \leq M$ for all (i, j) -entries a_{ij} of A . Then one can easily show that $[A^k]_{ij} \leq n^{k-1}M^k$ for all k and i, j . Thus

$$[e^A]_{ij} \leq \sum_{k=0}^{\infty} \frac{1}{k!} n^{k-1} M^k = \frac{1}{n} e^{nM},$$

so by the comparison test, each entry of $e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$ is absolutely convergent for any square matrix A . $\square$

Example 3.8.4. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$, then

$$\begin{aligned} e^A &= I + A + \frac{1}{2}A^2 + \dots \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 1+3 \\ 0 & 3^2 \end{bmatrix} + \dots = \begin{bmatrix} e^1 & * \\ 0 & e^3 \end{bmatrix}. \end{aligned}$$

It is a good exercise to calculate the missing entry $*$ directly from the definition. $\square$

Exercise 3.8.5. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Find $\lim_{k \rightarrow \infty} A^k$ if it exists. (Note that the matrix A is not diagonalizable.)

The following theorem is sometimes helpful to compute e^A , the proof is easy and we skip the proof as its application is far more significant than its proof.

Theorem 3.8.6. *Let $A_1, A_2, A_3, \dots$ be a sequence of $m \times n$ matrices such that $\lim_{k \rightarrow \infty} A_k = L$. Then*

$$\lim_{k \rightarrow \infty} BA_k = BL \quad \text{and} \quad \lim_{k \rightarrow \infty} A_k C = LC$$

for any matrices B and C for which the products can be defined.

For example, if A is a diagonalizable matrix and $Q^{-1}AQ = D$ is diagonal for some invertible matrix Q , then, for each integer $k \geq 0$, $A^k = QD^kQ^{-1}$ and

$$\lim_{k \rightarrow \infty} A^k = Q \left(\lim_{k \rightarrow \infty} D^k \right) Q^{-1} = Q \begin{bmatrix} \lim_{k \rightarrow \infty} \lambda_1^k & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \lim_{k \rightarrow \infty} \lambda_n^k \end{bmatrix} Q^{-1}.$$

Thus, $\lim_{k \rightarrow \infty} A^k$ exists if and only if $\lim_{k \rightarrow \infty} \lambda_i^k$ exists for $i = 1, 2, \dots, n$.

Also, by the above theorem,

$$\begin{aligned} e^A &= e^{QDQ^{-1}} = I + QDQ^{-1} + \frac{(QDQ^{-1})^2}{2!} + \frac{(QDQ^{-1})^3}{3!} + \dots \\ &= Q \left(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \dots \right) Q^{-1} \\ &= Qe^DQ^{-1}, \end{aligned}$$

whose computation is easy.

Example 3.8.7 (Computing e^A for a diagonalizable matrix A). Let $A = \frac{1}{2} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Then its eigenvalues are 1 and 2 with associated eigenvectors $\mathbf{u}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, respectively.

Thus $A = QDQ^{-1}$ with $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$. Therefore,

$$e^A = \frac{1}{2} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^1 & 0 \\ 0 & e^2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} e + e^2 & -e + e^2 \\ -e + e^2 & e + e^2 \end{bmatrix}. \quad \square$$

The following theorem shows some basic properties of the exponential matrices, whose proofs are easy, and are left for exercises.

Theorem 3.8.8. (1) $e^{A+B} = e^A e^B$ provided that $AB = BA$.
 (2) e^A is invertible for any square matrix A , and $(e^A)^{-1} = e^{-A}$.
 (3) $e^{Q^{-1}AQ} = Q^{-1}e^AQ$ for any invertible matrix Q .
 (4) If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of a matrix A with their associated eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then e^{λ_i} 's are the eigenvalues of e^A with the same associated eigenvectors $\mathbf{v}_i$'s for $i = 1, 2, \dots, n$. Moreover, $\det e^A = e^{\lambda_1} \dots e^{\lambda_n} = e^{\text{tr}(A)} \neq 0$ for any square matrix A .

In general, the computation of e^A is not easy at all if A is not diagonalizable. However, if A is a triangular matrix, it is relatively easy as shown in the following example.

Example 3.8.9 (Computing e^A for a triangular matrix of the form $A = \lambda I + N$). For $A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$, compute e^A .

Solution: Write $A = 2I + N$ with $N = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$. Since $(2I)N = N(2I)$, by Theorem 6.25(1), $e^A = e^{2I}e^N$. From the direct computation of the series expansion, we get $e^{2I} = e^2I$. Moreover,

since $N^k = \mathbf{0}$ for $k \geq 2$, $e^N = I + N + \frac{N^2}{2!} + \cdots = I + N = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. Thus,

$$e^A = e^2(I + N) = e^2 \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^2 & 3e^2 \\ 0 & e^2 \end{bmatrix}.$$

Exercise 3.8.10. Compute e^A for $A = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$.

3.9 Diagonalization of linear transformations

Definition 3.9.1. Let V be an n -dimensional vector space, and let $T : V \rightarrow V$ be a linear transformation on V . Then the **eigenvalues** and **eigenvectors** of T can be defined by the same equation, $T\mathbf{x} = \lambda\mathbf{x}$, with a nonzero vector $\mathbf{x} \in V$.

The following example illustrates how to find a diagonal matrix representation of a linear transformation on a vector space.

Example 3.9.2. Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the linear transformation defined by

$$(Tf)(x) = f(x) + xf'(x) + f'(x).$$

Find a basis for $P_2(\mathbb{R})$ with respect to which the matrix of T is diagonal.

Solution: First of all, we find the eigenvalues and the eigenvectors of T . Take a basis for the vector space $P_2(\mathbb{R})$, say $\alpha = \{1, x, x^2\}$. Then the matrix of T with respect to α is

$$[T]_{\alpha} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix},$$

which is upper triangular. Hence, the eigenvalues of T are $\lambda_1 = 1$, $\lambda_2 = 2$ and $\lambda_3 = 3$. By a simple computation, one can verify that the vectors $\mathbf{x}_1 = (1, 0, 0)$, $\mathbf{x}_2 = (1, 1, 0)$ and $\mathbf{x}_3 = (1, 2, 1)$ are eigenvectors of $[T]_{\alpha}$ in $\mathbb{R}^3$ belonging to eigenvalues λ_1 , λ_2 , λ_3 , respectively. Their associated eigenvectors of T in $P_2(\mathbb{R})$ are $f_1(x) = 1$, $f_2(x) = 1 + x$, $f_3(x) = 1 + 2x + x^2$, respectively. Since the eigenvalues λ_1 , λ_2 , λ_3 are all distinct, the eigenvectors $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ of $[T]_{\alpha}$ are linearly independent and so are $\beta = \{f_1, f_2, f_3\}$ in $P_2(\mathbb{R})$. Thus, each f_i is a basis for the eigenspace $E(\lambda_i)$ of T belonging to λ_i for $i = 1, 2, 3$, and the basis-change matrix is

$$Q = [id]_{\beta}^{\alpha} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3] = \begin{bmatrix} [f_1]_{\alpha} & [f_2]_{\alpha} & [f_3]_{\alpha} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Hence, by changing the basis α to β , the matrix representation of T is a diagonal matrix:

$$[T]_{\beta} = [id]_{\alpha}^{\beta} [T]_{\alpha} [id]_{\beta}^{\alpha} = Q^{-1} [T]_{\alpha} Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = D.$$

Exercise 3.9.3. Let T be the linear transformation on $\mathbb{R}^3$ defined by

$$T(x, y, z) = (4x + z, 2x + 3y + 2z, x + 4z).$$

Find all the eigenvalues and their eigenvectors of T and diagonalize T .